PTM PREDICTION WITH PROT-MAMBA

Harsh Chauhan hc3725@rit.edu

Advisor: **K.C. Dukka** dkcvcs@g.rit.edu

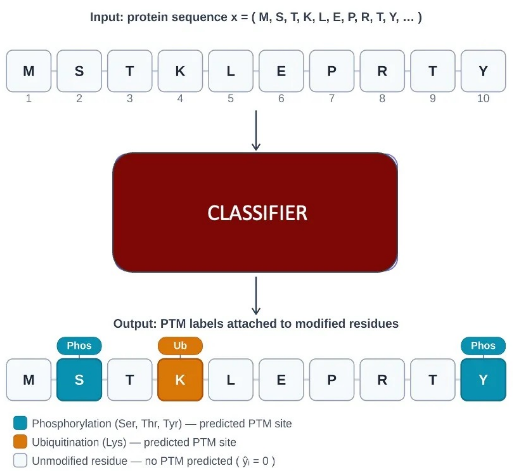
INTRODUCTION

Post Translation Modifications (PTMs) are biochemical mechanisms altering protein properties after synthesis, regulating critical biological processes.

Computational Challenge

- Wet lab experiments has detection limitations
- Transient modifications at substoichiometric levels
- Sampling biases in enrichment methods

Solution: Machine learning models for rapid, scalable PTM site prediction across entire proteomes.



METHODOLOGY

We approach PTM site prediction as a supervised classification problem on residue-centered sequence windows, where each candidate site is represented in a consistent, fixed-length format to enable fair comparisons across PTM types and modeling choices. Our methodology separates the pipeline into two stages: (i) representation learning via PTM-Mamba feature extraction to capture informative sequence context efficiently, and (ii) downstream classifier training to map those representations (or simpler baseline encodings) to PTM/non-PTM labels under class imbalance. This design lets us reuse the same extracted embeddings across multiple architectures, iterate quickly, and evaluate per-PTM and cross-PTM generalization in a controlled way.

CNN + RNN ARCHITECTURE

WHY MAMBA

We use Mamba-based representations because PTM prediction is fundamentally a sequence-context problem: whether a site is modified depends on **patterns** in the **local window** and its broader biochemical context. Mamba provides a strong, efficient sequence backbone that can encode informative **residue dependencies** into a **fixed-size embedding**, giving downstream classifiers a richer signal than sparse one-hot features while remaining practical to run at scale. By extracting PTM-Mamba hidden states once and reusing them across tasks, we decouple expensive representation learning from lightweight classifier training, enabling fast iteration across PTM types and model variants

MODELS

Baseline MLP	CNN+GRU	MultiClassifier	ptmgpt2
Window + entropy features	PTM-Mamba embeddings	All PTMs, conditioned	External benchmark

We classify PTM sites using a residue-centered window pipeline and train three complementary models: (1) a baseline MLP on fixed window features to establish a strong reference, (2) a CNN+GRU sequence model that consumes PTM-Mamba embeddings to capture richer local context, and (3) a pooled multi-PTM classifier (optionally conditioned on PTM type) to learn shared patterns across modification types while retaining PTM-specific decision boundaries. All models are evaluated on held-out test splits with imbalance-aware metrics (e.g., MCC/F1/recall), and we additionally compare against (4) PTMGPT2 as an external benchmark to contextualize performance.

RESULTS

MODEL COMPARISON: F1 SCORE

Fig 1. CNN+GRU outperforms MLP. MultiClassifier is competitive across all PTMs.

MODEL COMPARISON: RECALL

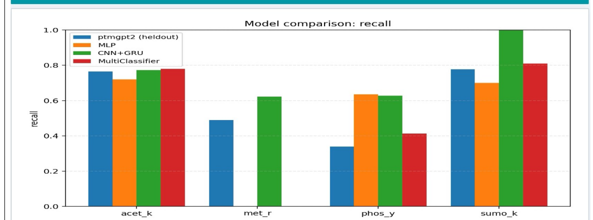


Fig 2. MultiClassifier matches or exceeds CNN+GRU recall on acet_k and sumo_k.

KEY RESULTS (TEST SET)

PTM	Model	MCC	F1	AUROC	AUPR
acet_k	CNN+GRU	0.502	0.739	—	0.750
acet_k	MultiClassifier	0.310	0.695	0.719	0.713
phos_y	MLP baseline	0.325	0.460	—	0.439
phos_y	MultiClassifier	0.332	0.441	0.791	0.454
sumo_k	MultiClassifier	0.339	0.662	0.735	0.654
met_r	MultiClassifier	0.335	0.481	0.751	0.501

CONCLUSION

Combining **PTM-Mamba embeddings** with **hybrid CNN+GRU** classifiers creates a robust, scalable PTM prediction pipeline.

Future work includes to train the models on remaining PTM datasets and finetune the multi-classifier model to increase upon the accuracy.

RIT
Rochester Institute of Technology

Computer Science @ RIT
B. Thomas Golisano College of Computing & Information Sciences